

Methodological Foundations of the ITHACA Probabilistic P-E Ensemble

Implementation baseline and overall assessment

The first important finding is that the **current code, not the older methodology notes, must be used to describe ITHACA**. The repository has evolved materially beyond `methodology_long_v1.1.md` and `figures_plan.md`. The implemented workflow now contains **ten weighting scenarios**, not eight; the all-scenario sensitivity experiment uses **100 Monte Carlo members per scenario**; a separate **1000-member change-weighted ensemble** is generated for the companion analysis; and `04c_create_gridded_mc_ensemble.R` is no longer empty but writes 100 BASE and 100 CHANGE gridded NetCDF realizations with explicit source-dataset provenance. By contrast, the older planning documentation still describes eight scenarios and an unimplemented `04c`, so those documents should not be allowed to propagate into the manuscript.

The implemented scientific architecture is internally coherent. Candidate P and E products are first evaluated separately against candidate-specific, leave-one-out multi-product reference ensembles. At every grid cell, ITHACA computes candidate and reference climatological means, interannual standard deviations, Sen slopes, and Mann-Kendall significance states. It then applies physical-plausibility gates, converts performance diagnostics into relative weights, aggregates grid-cell weights to area-weighted **IPCC AR6 region × biome** probabilities, and samples one dataset identity per region × biome and ensemble member. That identity is retained for the full 1982–2021 time series and supplies both P and E for the sampled “world.”

The high-level methodological verdict is therefore:

ITHACA is scientifically defensible as an explicit Monte Carlo representation of structural uncertainty associated with dataset choice, conditional on its candidate set, reference ensemble, plausibility gates, spatial stratification, and weighting rule. It should not be presented as a statistically calibrated posterior distribution of the true P and E fields, nor should its ensemble spread be described as total observational uncertainty.

That distinction is crucial. In a Bayesian model-averaging framework, mixture weights are posterior model probabilities derived from an explicit likelihood and training data; ITHACA instead constructs normalized skill/plausibility scores and uses them as categorical sampling probabilities. Both define valid probability distributions mathematically, but they have different epistemic meanings. ¹

There are also several issues that deserve action before publication rather than merely additional prose. The most consequential are the treatment of trend significance and autocorrelation, the ill-conditioning of relative slope error when the reference trend is close to zero, residual dependence among products in the consensus reference, the sensitivity of the two physical gates, and a code/text discrepancy concerning

trend-sign agreement. None invalidates the framework, but several could materially change the weights and therefore need either correction or demonstrated robustness. 2

Evidence audit across the methodological components

ITHACA choice	Supporting literature	Challenging literature	Main assumptions	Main limitation	Recommended manuscript justification
Multi-product reference evaluation: candidate-specific leave-one-out median reference for P and E, with medians of product means, SDs and Sen slopes rather than a single “truth” dataset.	Collocation literature starts from exactly the problem ITHACA faces: nominal reference datasets are themselves uncertain. Triple collocation estimates error characteristics without designating one system as truth; Extended Triple Collocation adds correlation to an unknown target. 3 Global P and ET work has subsequently used collocation and error-covariance methods for multi-product evaluation and merging. 4	Classic TC assumes mutually independent errors and a linear error model; violations can bias estimated errors. Yilmaz & Crow explicitly tested these assumptions, while SUPER and later collocation methods were developed partly because inter-product error correlation cannot safely be ignored. 5 A product median is therefore a consensus , not an independent truth.	Most products contain useful information; errors are not so strongly genealogically correlated that the median is systematically wrong; leave-one-out removal sufficiently limits direct self-validation.	LOO removes a candidate's direct self-contribution but not common forcing, assimilated observations, retrieval heritage, model structure, or correction datasets shared among products. Several closely related products can therefore dominate consensus.	State that ITHACA uses a robust multi-product consensus surrogate in the absence of a global truth , with candidate-specific LOO to prevent direct self-inclusion. Explicitly say that this does not eliminate cross-product dependence . Add median-vs-mean and preferably family/dependency sensitivity; collocation should be presented as an important complementary approach rather than something the current method supersedes.

ITHACA choice	Supporting literature	Challenging literature	Main assumptions	Main limitation	Recommended manuscript justification
Separate evaluation of climatological mean, temporal variability, trend magnitude and trend significance. Mean and SD use relative absolute error; trend uses Sen slope; significance uses Mann–Kendall state.	Multidimensional evaluation is standard because no single statistic captures bias, variance and temporal correspondence. Taylor explicitly combines complementary statistics, while Gleckler et al. developed multidimensional performance metrics for model intercomparison. ⁶ Sen's estimator is a robust rank-based slope estimator. ⁷	Standard Mann–Kendall assumes independent observations. Hamed & Rao showed that autocorrelation requires modified inference; otherwise significance can be distorted. ⁸ In ITHACA this matters unusually strongly because the <code>change</code> scenario gives trend-significance agreement 75% of the change component, while change receives 75% of the climatology/change split.	Climatological fidelity, temporal variability and long-term change provide distinct evidence about dataset behavior; the consensus trend is sufficiently meaningful to score candidates.	The significance component can be highly sensitive to serial correlation and to the arbitrary α threshold. The code uses p < 0.1, and the current categorical significance test does not actually compare the candidate's slope sign with the stored <code>majority_sign</code>.	Strongly justify separate climatology and change diagnostics, but do not claim that present-day mean skill guarantees trend skill. Use autocorrelation-aware trend inference, or at minimum repeat weighting with modified MK / alternative α. Correct manuscript language from “trend sign and significance agreement” to what is actually implemented—or change the code to test sign explicitly.

ITHACA choice	Supporting literature	Challenging literature	Main assumptions	Main limitation	Recommended manuscript justification
Spatially varying performance, evaluated at grid cells but regularized to region × biome sampling probabilities. The regional probability uses an area-weighted average over all retained cells; dataset absence contributes implicitly zero.	Global ET intercomparison shows that different retrieval systems perform differently across ecosystems, climatic regimes and water-stress conditions; WACMOS-ET specifically proposed biome/ climate-specific composites and performance-weighted ensembles. ⁹ DOLCE implements spatial/temporal tiering and explicitly explores grouped weighting. ¹⁰ OpenET now maps model skill as a function of climate and land cover and derives spatially varying weights. ¹¹	Finer stratification is not automatically better. DOLCE found that climate/land-cover grouping could fail when partitioning reduced observational sample sizes and promoted overfitting; later versions deliberately chose broader groups or regimes with sufficient support. ¹²	Region × biome units are internally similar enough that one source choice can represent them, while being coarse enough to suppress noisy cell-to-cell switching.	Hard unit boundaries can introduce artificial discontinuities. Some region×biome combinations are small. Moreover, the all-cell denominator means the regional probability represents skill × spatial availability/ plausibility, not conditional skill alone.	Present region×biome aggregation as spatial regularization of locally evaluated skill chosen to trade local adaptivity against spatial coherence—not as an objectively optimal partition. Quantify cell count/area per unit. Compare at least region-only or biome-only weighting for sensitivity. Explicitly say that the all-cell denominator intentionally penalizes spatially patchy eligibility.

ITHACA choice	Supporting literature	Challenging literature	Main assumptions	Main limitation	Recommended manuscript justification
Performance-to-weight transformation: normalized inverse loss for the primary hierarchy; rank-linear, rank-exponential, neutral and inverted alternatives.	Performance weighting has a long precedent, including REA's performance/convergence weights and Bayesian predictive weighting. ¹³ Knutti et al. formalized weighting based jointly on observational performance and model dependence. ¹⁴ OpenET tested multiple metric-to-weight mappings and explicitly varied weighting strength. ¹¹	Weighting can make an ensemble worse when estimated skill is noisy or unrelated to the target quantity. Weigel et al. showed that incorrectly specified performance weights can underperform equal weighting, making equal weighting an important benchmark. ¹⁵ There is no general theorem making inverse error the uniquely correct transform.	Smaller disagreement with the chosen reference is evidence for greater relative credibility; the loss-to-weight function is monotonic and useful over the relevant loss range.	$1/(\epsilon + \text{loss})$ has an arbitrary scale/aggressiveness. In particular, the slope loss $s_i - s_{\text{ref}} / (s_{\text{ref}} + 10^{-6})$ becomes ill-conditioned when the consensus slope is near zero. The resulting numbers are not calibrated posterior truth probabilities.	Call them selection probabilities derived from normalized performance scores, or relative sampling weights, not "probability that product i is correct." Keep neutral weighting as the principal baseline. Test weighting strength, ideally $w_i \propto (\epsilon + L_i)^{-\alpha}$ over several α values in addition to the existing rank and adversarial scenarios.

ITHACA choice	Supporting literature	Challenging literature	Main assumptions	Main limitation	Recommended manuscript justification
Physical plausibility screening: $P/E \geq 0.9$ and AET below PET in a strict majority of 12 PET comparisons (<code>n_below_pet > 6</code>).	Budyko theory gives a strong first-order water-energy constraint on long-term catchment-scale P, E and available energy/PET. ¹⁶ Water-budget fusion studies likewise show the value of hydrological closure constraints in multi-source estimation. ¹⁷	$E \leq P$ is not a universal hard constraint. Standard Budyko assumes near-steady storage; Greve et al. explicitly generalized Budyko to represent $E > P$ when additional water is available from storage or other sources. ¹⁸ Likewise PET depends on its definition and forcing, so $AET \leq PET$ should not be treated as an exact observational law at every grid cell.	Multiannual P/E and PET consistency are informative screens for grossly implausible combinations; a majority vote across PET estimates reduces dependence on one PET product.	Both thresholds are heuristic. $P/E \geq 0.9$ permits E to be about 11% greater than P but can still reject physically legitimate irrigated, groundwater-fed, storage-depleting or laterally supplied regions. The 7-of-12 PET threshold has no unique physical derivation.	Use the term physical-plausibility gates, not “physical constraints.” Explain that they reject gross inconsistencies while allowing modest $E > P$. Show sensitivity to P/E threshold and PET-vote threshold and map exclusions by reason. Treat basin water-budget closure with runoff/storage as an independent validation, not proof that these local gates are exact.

ITHACA choice	Supporting literature	Challenging literature	Main assumptions	Main limitation	Recommended manuscript justification
Joint P-E treatment: sample one candidate identity and carry its paired P and E together for all years in a region×biome realization. The GLEAM world deliberately uses MSWEP P with GLEAM E; the other candidate worlds are paired analysis systems.	Multivariate water-budget fusion literature argues for treating P, E, runoff and storage coherently rather than correcting each independently. Pan et al. combine multiple fluxes under a water-balance constraint, while Aires demonstrates multivariate integration and closure methods. 17 More recent Bayesian water-budget work similarly treats flux errors jointly rather than as isolated scalar uncertainties. 19	The chosen P-E pair is more physically/statistically coherent than an arbitrary combination of independently selected P and E products; pair covariance is meaningful enough to preserve.	Pairing can under-explore marginal P and E uncertainty because only five joint worlds are accessible rather than the full P×E product space. “Coherent” also has different meanings: MSWEP-GLEAM is an intentionally constructed pair, not P and E produced by the same land model.	Defend pairing as preservation of joint source provenance and P-E covariance, not as proof of perfect water-budget closure. State explicitly how MSWEP-GLEAM differs from same-system pairs. Include an independent-P/E sampling experiment or all-combinations sensitivity to show how much pairing constrains P-E uncertainty.	

ITHACA choice	Supporting literature	Challenging literature	Main assumptions	Main limitation	Recommended manuscript justification
Monte Carlo propagation of dataset-choice uncertainty: categorical inverse-CDF sampling of dataset identity per simulation $\times$ region $\times$ biome, with one source retained through the full time series and recorded in <code>source_dataset_id</code>. The same uniform draw is reused across weighting scenarios.	Observational ensemble datasets such as E-OBS establish the principle that observational uncertainty can be represented through stochastic ensemble realizations rather than a single deterministic field. ²⁰ Water-budget work has also generated ensembles of alternative realizations from combinations of source datasets and closure choices rather than asserting one “correct” combination. ²¹ Common random numbers are an established variance-reduction design for paired comparisons of alternative stochastic systems. ²²	The candidate set represents the relevant structural choices; keeping identity fixed through time is preferable to creating artificial year-to-year source switching; region$\times$biome independence is an adequate approximation to cross-regional dependence.	This propagates dataset-choice uncertainty only. It does not generate each source product's measurement/retrieval uncertainty. With only 100 public/sensitivity members, tail quantiles and rare-event frequencies can remain Monte-Carlo noisy unless convergence is demonstrated.	Define the ensemble as a provenance-preserving structural ensemble. Emphasize that MC propagates uncertainty in dataset identity through nonlinear derived quantities while preserving each source's temporal structure. Provide convergence diagnostics against the known analytical weighted expectation. Do not imply that member spread is complete observational error.	

ITHACA choice	Supporting literature	Challenging literature	Main assumptions	Main limitation	Recommended manuscript justification
Alternative weighting scenarios: base, change, trend-dominant, climatology-dominant, P-dominant, E-dominant, rank-linear, rank-exponential, neutral and inverted. Scenarios share MC draws.	The weighting literature strongly supports testing equal weighting and dependence on weighting strength/diagnostics rather than treating one weighting scheme as self-evidently correct. ²³ OpenET's explicit exploration of seven metric-to-weight mappings and different weighting powers is particularly close methodologically. ¹¹	An adversarial inverted scenario is not a plausible posterior alternative; it deliberately emphasizes disagreement. Common-random-number comparisons also create a coupling between categorical distributions that depends on how categories are ordered.	The selected scenarios span the important structural dimensions of the weighting rule sufficiently to reveal sensitivity.	Ten scenarios still do not exhaust reference uncertainty, dependence assumptions, physical-gate thresholds or spatial-unit uncertainty. Some scenarios are stress tests, not competing scientific estimates.	Present scenarios in three semantic classes: reference/primary, plausible methodological alternatives, and stress-test benchmarks. Explicitly identify neutral as the no-skill benchmark and inverted as adversarial. Add reference-composition, gate-threshold and weighting-strength sensitivities rather than interpreting the existing ten as a complete uncertainty distribution.

Cross-cutting methodological issues that matter most

Consensus without truth is defensible, but dependence must become visible

ITHACA's leave-one-out construction solves a real circularity problem: when a candidate is contained in the reference ensemble, it is removed before that candidate is evaluated. The code explicitly records whether removal occurred, and for the GLEAM/MSWEP pairing the precipitation and evaporation cases are treated differently according to actual membership in the corresponding reference ensemble. This is appreciably better than scoring every product against an ensemble average that includes itself.

It does **not**, however, make the reference independent. Triple-collocation research is particularly instructive here. TC works precisely because the “truth” is latent, but classic forms require errors in the measurement systems to be mutually uncorrelated. ³ Yilmaz and Crow showed why violations of TC assumptions

matter, and newer precipitation-merging methods such as SUPER explicitly estimate error covariances because ignoring inter-product dependence can lead to overconfident fusion. ⁵ ET collocation studies reach the same conclusion: cross-error correlation is a real quantity that needs consideration, not a theoretical curiosity. ²⁴

For ITHACA, this means that “agreement with the reference ensemble” should never be translated into “accuracy relative to observations.” The scientifically defensible term is closer to **consistency with the multi-product consensus**.

A particularly valuable additional analysis would be **leave-one-family-out**, not just leave-one-product-out. For example, products sharing forcing, assimilation systems, model ancestry, gauge inputs or remote-sensing retrievals could be grouped; each family could then be removed in turn. This directly tests whether high weights reflect multiple independent lines of evidence or simply replicated methodology. That recommendation follows from the dependence problem established in collocation and model-ensemble literature. ²⁵

Mean-state skill and change skill should remain separate

The decision not to collapse climatological state, variability and change into one raw error metric is methodologically strong. Classical performance frameworks treat correlation, variance and error magnitude as distinct properties, while multimodel evaluation shows that “good performance” depends on what aspect of the system is being evaluated. ⁶

For ITHACA this is more than statistical nicety. A product can have a realistic 1982–2021 climatological mean after bias correction while inheriting an incorrect trend from changing satellite inputs, reanalysis observing-system changes or forcing data; conversely, a product with a mean bias can still capture temporal anomalies or long-term change. The manuscript should therefore defend separate climatology/change components as **orthogonal evidence about fitness for purpose**, not as an attempt to produce one universal ranking. The strength assigned to those components is then an explicit scientific preference explored through scenarios.

The main weakness is the current significance metric.

The reference calculation stores `majority_sign`, but `03a_dataset_ranking.R` does not use that sign when constructing `*_check_significance`. If the reference ensemble has a significant positive majority, a candidate with a significant **negative** trend can still receive `check_significance = TRUE`; its opposite sign is punished only through the continuous slope-difference component. Consequently, manuscript wording such as “agreement with the reference majority on trend significance **and sign**” would currently be incorrect.

There is a second issue. `trend::mk.test` is used directly with `p < 0.1`. Classical Mann–Kendall inference assumes independence, whereas Hamed and Rao developed a modified test specifically because autocorrelation alters the null variance and therefore trend significance. ⁸ This would be a minor detail if significance were merely descriptive. It is not minor in the `change` weighting scenario: change receives 75% of the climatology/change blend and significance receives 75% of the change subcomponent.

I would therefore treat an **autocorrelation-aware significance sensitivity** as close to mandatory. Suitable options include modified MK, block/bootstrap significance, effective-sample-size adjustment, or simply reducing the categorical significance component and demonstrating that results are insensitive to its removal. Changing α from 0.10 to 0.05 should also be included in that test.

A “sampling probability” is legitimate; a “probability of truth” is not

ITHACA's weights are transformed into positive numbers summing to one and are then sampled categorically. Mathematically, they are therefore probabilities. The issue is **what event those probabilities describe**.

Bayesian model averaging provides a useful contrast: BMA weights are posterior probabilities under an explicitly specified predictive model and training likelihood. ²⁶ ITHACA does not estimate that object. Its inverse-loss weights encode a scientific rule of the form “products agreeing more closely with our chosen evidence receive greater probability of being selected.”

The manuscript can safely say:

“The normalized weights define dataset-selection probabilities conditional on the evaluation metrics, reference ensemble, physical-plausibility screening and chosen weighting scenario.”

It should avoid formulations such as:

“Dataset i has a 40% probability of being the true dataset.”

The difference also determines what the ensemble uncertainty means. The ensemble spread is an estimate of **conditional structural uncertainty from plausible source-dataset choice**. It excludes uncertainty shared by all candidate products, uncertainty outside the candidate envelope, within-product retrieval uncertainty unless already represented in the products, uncertainty in the reference construction, and uncertainty in the methodological hyperparameters unless those are explicitly integrated over.

This wording is not a weakness. It makes the uncertainty target scientifically identifiable.

The physical filters are useful safeguards, not laws of nature

The hydrological motivation for both gates is sound. At long averaging scales, Budyko theory relates precipitation, evaporation and available energy, and water-budget fusion methods have demonstrated that enforcing hydrological consistency can improve multi-source estimates. ²⁷

But the ITHACA gates should remain explicitly **soft plausibility criteria**.

The P/E gate rejects $P/E < 0.9$, equivalent to permitting E up to roughly $1/0.9 \approx 1.11P$. That tolerance is actually a sensible sign that the implementation does not assume exact $E \leq P$. Yet Greve et al. show formally why $E > P$ can be hydrologically legitimate whenever evapotranspiration draws on water beyond contemporaneous precipitation; traditional Budyko limits depend on steady-state assumptions regarding storage. ¹⁸

Likewise, requiring AET to lie below PET for at least 7 of 12 PET estimates is an appealing consensus screen, but the threshold is a design choice rather than an immutable physical constant.

For an ESSD paper, the strongest treatment is not to defend 0.9 or 7/12 rhetorically. It is to show that conclusions do not hinge on them. A compact threshold experiment—e.g. several P/E tolerances and PET-majority requirements—could report retained land area, dominant-dataset changes, effective number of datasets, and changes in ensemble P/E summaries. This would turn an obvious reviewer vulnerability into a strength.

Sensitivity and validation priorities

The existing ten-scenario architecture is already substantially better than presenting one supposedly optimal weight rule. The current code contains a balanced base case, a change-dominant configuration, pure trend and pure climatology endpoints, P-only and E-only endpoints, two rank transforms, exact neutral weighting and an intentionally inverted adversarial case. That design is consistent with the broader ensemble literature, where skill weighting may help but can also underperform equal weighting when skill estimates are noisy or misspecified. ²⁸

What the current scenarios do **not** fully test are the assumptions upstream of the weighting transform. The priority sensitivity matrix should therefore be broader than “ten weighting scenarios.”

Priority test	Why it matters	Minimal defensible implementation
Reference construction	Median consensus may inherit family/common-mode bias; LOO only removes direct self-inclusion. ⁵	Compare LOO median with LOO mean; add leave-one-family-out where provenance permits.
Trend significance	Unmodified MK with $p < 0.1$ receives high nominal influence in change weighting. ⁸	Repeat with autocorrelation-corrected MK and $p < 0.05$; optionally remove the categorical significance component.
Trend sign	Current categorical significance test does not compare candidate sign with <code>majority_sign</code> .	Either implement explicit sign agreement or remove “sign agreement” from methods and quantify the consequence.
Near-zero slopes	Relative slope error is normalized by $ s_{ref} + 10^{-6}$, making the slope component sensitive when consensus trend is close to zero.	Compare against absolute slope difference scaled by reference IQR, interannual variability, or a fixed physically interpretable trend scale.
Weight aggressiveness	Inverse-error weighting has no uniquely correct curvature, and overly aggressive weights can overfit noisy skill. ²⁸	Add an exponent α to inverse loss and compare $\alpha = 0$ (uniform), weak, present and stronger weighting.

Priority test	Why it matters	Minimal defensible implementation
Physical gates	Thresholds 0.9 and 7/12 are heuristics; legitimate E>P cases exist. ¹⁸	Threshold grid; report coverage, source dominance and ensemble-summary changes.
Spatial stratification	Local adaptation helps, but excessive grouping can fragment evidence and overfit. ²⁹	Compare region×biome against region-only and, if affordable, biome-only or a hydroclimatic-regime alternative.
Regional denominator	Current denominator penalizes products that fail the physical gate over part of a unit.	Compare all-cell denominator with conditional-on-surviving-footprint averaging; show which interpretation matters.
P-E pairing	Pairing preserves covariance but excludes many possible P×E combinations.	Compare paired sampling with independent marginal P/E sampling as a diagnostic, not necessarily as the released product.
MC convergence	The public ensemble has 100 members although exact categorical probabilities are known upstream.	Plot convergence of means, medians, 5/95% quantiles and storyline fractions versus N; benchmark means against analytic weighted expectations.
Candidate-set dependence	No weighting system can represent plausible structural worlds absent from the candidate set; product composition can matter as much as the fusion algorithm.	Leave-one-candidate-out ensemble diagnostics; report changes in central estimates and uncertainty.

One additional subtlety deserves attention if the manuscript uses **member-by-member differences between scenarios**. ^{04a} deliberately passes the same uniform variate u through every scenario's categorical CDF, after fixing a global alphabetical/deterministic order of datasets. Common random numbers are well established for reducing Monte Carlo noise in comparisons, so the basic strategy is good.

²²

For categorical distributions, however, inverse-CDF coupling is not invariant to the arbitrary ordering of categories. Reordering the five dataset labels leaves every scenario's **marginal** categorical probabilities unchanged but can change which source is paired with which source under the same u . That means scenario-average comparisons are sound, but statistics such as “fraction of paired members whose dataset identity flips from scenario A to B” can depend on category ordering. This is a mathematical consequence of the implemented CDF coupling. If such paired provenance-switch metrics become an important result, either test several category orders or use an order-invariant/maximal categorical coupling. If the common draws are used only to reduce stochastic noise in aggregate scenario contrasts, this issue is much less consequential.

The current MC construction has another important virtue that should be highlighted: **dataset identity is fixed through time** within a sampled region×biome world. This preserves the complete source time series—its autocorrelation, covariance between wet and dry years, trend structure and P-E relationship. Sampling dataset identity independently every year would instead manufacture a temporally discontinuous pseudo-record whose trend and variance no input product actually possesses. The present design is therefore particularly well suited to the intended analysis of multidecadal change.

Likewise, preserving `source_dataset_id` in every gridded member is scientifically valuable because the ensemble does not erase provenance when uncertainty is propagated. That distinction separates ITHACA conceptually from a conventional weighted-mean fusion product: an ensemble member remains traceable to its component source choices.

Publication assessment

Methods strongly supported by literature

Separating mean state, variability and temporal change. This is firmly defensible. Multidimensional model/data evaluation has long recognized that performance is property-dependent and should not be collapsed prematurely into one statistic. ⁶ Sen's slope is also an established robust trend estimator. ⁷

Spatially varying dataset performance. This has strong direct support in the P/ET literature. WACMOS-ET found product performance to vary across ecosystem and climate conditions and explicitly argued for condition-dependent composites; DOLCE and OpenET subsequently implemented increasingly sophisticated spatial/regime-dependent weighting strategies. ³⁰

Using multiple imperfect products rather than declaring one global product to be truth. That premise is fundamental to triple/extended collocation and multi-source fusion. ³¹

Leave-one-out removal of the evaluated product from its own consensus reference. This is a straightforward and scientifically sound protection against the most immediate form of circularity. It does not solve product dependence, but it is clearly preferable to self-inclusive scoring.

Preserving P-E pairing and full source time series within an MC member. This is defensible for an ensemble intended to represent coherent multivariable structural alternatives. Multivariate water-cycle fusion literature gives strong reason not to treat hydrological variables as unrelated scalar quantities. ¹⁷

Keeping equal weighting as a benchmark and testing multiple weighting rules. This is exceptionally well justified by the literature because performance weighting is useful only when the performance information itself is reliable. ³²

Monte Carlo propagation with explicit provenance. Ensemble observations are an established way of representing data uncertainty, while ITHACA's categorical source sampling specifically propagates structural source choice. ³³ The fixed source ID makes the resulting uncertainty auditable rather than anonymous.

Reasonable pragmatic choices

The leave-one-out ensemble median as the working reference is pragmatic and defensible where no homogeneous global truth exists. It gains robustness against isolated outlying products but cannot protect against common-mode error. It should therefore be called a consensus reference rather than observational truth. ³⁴

Region × biome aggregation is a sensible regularization strategy. It avoids the unrealistic salt-and-pepper field that would result from independent grid-cell selection while allowing considerably more spatial adaptability than a single global ranking. The exact partition is not uniquely justified by theory, but the principle is strongly supported. ²⁹

Inverse-loss weighting is a legitimate monotone scoring rule, but not a uniquely optimal one. That places it squarely in the category of pragmatic ensemble design rather than statistical calibration. The existing rank, uniform and adversarial alternatives make that choice substantially easier to defend. ³⁵

The all-cell regional denominator is reasonable if the intended quantity is “suitability for representing this entire region×biome unit,” because products failing the physical filter across large portions of the unit should then be penalized. It is not neutral: it deliberately mixes conditional skill and spatial eligibility.

The P/E and multi-PET gates are reasonable quality-control devices provided they are explicitly characterized as empirical plausibility filters and subjected to threshold sensitivity. Budyko and water-budget theory support the physical direction of the tests, not those exact numerical thresholds. ³⁶

Choices requiring explicit sensitivity analysis

The highest-priority sensitivity analyses are, in descending order:

Trend significance and autocorrelation. The combination of unmodified MK, $p < 0.1$ and high significance weight in the change-dominant configuration creates a realistic path by which serial correlation could alter regional dataset probabilities. ⁸

Reference dependence and genealogy. LOO is insufficient against correlated product families. A family-aware exclusion test would address this directly. ⁵

Near-zero trend normalization. The present relative slope loss can assign very different slope scores to tiny absolute differences when the consensus slope is close to zero. A scale based on ensemble IQR or a physically meaningful trend unit would be less ill-conditioned.

Physical gate thresholds. P/E=0.9 and 7/12 PET agreement need sensitivity because neither is a hard physical boundary. ¹⁸

Weighting strength. Existing scenarios vary the ingredients and transformation, but a direct continuum from uniform to aggressive inverse-loss weighting would make the robustness argument substantially stronger. OpenET's explicit exploration of weighting powers provides a close precedent. ¹¹

P-E coupling. The released paired ensemble can remain the preferred product, but independent P/E sampling should be shown as a diagnostic to quantify uncertainty suppressed—or unrealistic combinations avoided—by pairing.

MC ensemble size. One hundred members may be adequate for many bulk statistics, but this should be demonstrated rather than asserted. The known analytical weighted mean provides an unusually convenient convergence benchmark.

Choices requiring careful wording rather than stronger validation

The following are primarily interpretation problems:

“Probability.” Use *selection probability, sampling probability, or relative plausibility weight*. Do not use *posterior probability of correctness*. ²⁶

“Uncertainty.” Use *dataset-choice uncertainty, between-product structural uncertainty, or uncertainty conditional on the candidate ensemble*. Do not imply full observational uncertainty.

“Physical constraint.” Prefer *physical-plausibility screen*. Budyko and PET considerations are strong diagnostic guidance, but storage, lateral supply and the definition of PET prevent these cell-level inequalities from being universal hard laws. ³⁶

“Consensus reference.” Do not call the LOO median “truth,” “observations,” or an “independent reference.” The products are themselves imperfect and partly dependent. ³⁷

“Coherent P-E dataset.” For ERA5-Land/MERRA-2/etc., source-system coherence is intelligible; for the GLEAM world, P is intentionally supplied by MSWEP while E comes from GLEAM. A safer generic expression is **“predefined P-E analysis pairs whose joint provenance is preserved during sampling.”**

“Trend sign agreement.” The current code does not implement a categorical candidate-vs-reference sign test. Until that changes, use *trend significance-state agreement plus continuous slope agreement*.

Issues that could threaten publication

The overall framework itself is not the publication threat. The threats are correctable implementation/validation issues.

The strongest scientific vulnerability is trend weighting. A reviewer could reasonably object that unmodified Mann-Kendall significance at $\alpha=0.10$ receives very large weight without addressing serial correlation. Hamed & Rao is a direct methodological challenge that will be difficult to dismiss. ⁸ Because significance has high leverage in the `change` scenario, this should be fixed or explicitly stress-tested before submission.

The sign-agreement code/text mismatch is a reproducibility issue. The reference script calculates `majority_sign`, but the candidate scoring logic checks whether a trend is significant or non-significant,

not whether its sign agrees with that majority. If the paper says otherwise, this is a concrete methods error rather than a matter of interpretation.

Residual product dependence is the major conceptual vulnerability. Calling LOO consensus an “independent” validation would invite strong criticism because modern fusion/collocation literature specifically emphasizes correlated source errors. ³⁸ The solution is not necessarily to abandon consensus weighting; it is to quantify dependency sensitivity and narrow the claim.

Hard filtering can induce structured missingness. Because eligibility depends on P/E and PET consistency, spatial holes are not random and can be product- and climate-dependent. The manuscript already plans to expose these masks, which is exactly the right response. But the threshold sensitivities should accompany the map.

Repository documentation currently disagrees with the source code on central dataset-paper facts. Current code implements ten scenarios and a populated public gridded MC generator, whereas older documentation describes eight scenarios and an empty `04c`. For an ESSD reproducibility paper, this needs to be reconciled before submission.

The practical publication judgment is therefore:

No methodological component makes ITHACA intrinsically indefensible. The design is publishable as a structural observational ensemble, provided the paper demonstrates sensitivity to the high-leverage subjective choices and does not over-interpret its weights. Trend-significance weighting and the current trend-sign implementation are the two items I would resolve before treating the methodology as submission-ready.

Essential methodological reference base

The following 25 papers form a compact core citation set for the Methods and Discussion. They are not simply the newest papers; each addresses a methodological assumption that ITHACA actually makes.

Reference	Core methodological role for ITHACA
Stoffelen (1998), “Toward the true near-surface wind speed: Error modeling and calibration using triple collocation,” JGR Oceans. ³⁹	Foundational argument for evaluating multiple imperfect geophysical observing systems without assuming one is error-free.
McColl et al. (2014), “Extended triple collocation: Estimating errors and correlation coefficients with respect to an unknown target,” GRL. ⁴⁰	Makes the unknown-truth problem explicit and shows that complementary error metrics are required.
Yilmaz & Crow (2014), “Evaluation of Assumptions in Soil Moisture Triple Collocation Analysis,” Journal of Hydrometeorology. ⁴¹	Essential caution on correlated errors and violated independence assumptions.

Reference	Core methodological role for ITHACA
Dong et al. (2022), “Statistical uncertainty analysis-based precipitation merging (SUPER),” Remote Sensing of Environment. 42	Direct precedent for multi-product precipitation weighting that explicitly accounts for inter-product error covariance.
Li et al. (2022), “Error characterization of global land evapotranspiration products: Collocation-based approach,” Journal of Hydrology. 24	Direct global-ET evidence that collocation and error cross-correlation are practical alternatives to a single reference.
Taylor (2001), “Summarizing multiple aspects of model performance in a single diagram,” JGR Atmospheres. 43	Foundational justification for multidimensional rather than single-metric evaluation.
Gleckler et al. (2008), “Performance metrics for climate models,” JGR Atmospheres. 44	Supports evaluating distinct aspects of system behavior and cautions against universal scalar skill.
Sen (1968), “Estimates of the Regression Coefficient Based on Kendall’s Tau,” JASA. 7	Statistical foundation for the robust Sen slope used by ITHACA.
Hamed & Rao (1998), “A modified Mann–Kendall trend test for autocorrelated data,” Journal of Hydrology. 8	Essential challenge/reference for ITHACA's trend-significance calculation.
Miralles et al. (2016), “The WACMOS-ET project – Part 2: Evaluation of global terrestrial evaporation data sets,” HESS. 9	Direct evidence for spatially varying ET performance and biome/climate-specific ensemble thinking.
Hobeichi et al. (2018), “Derived Optimal Linear Combination Evapotranspiration (DOLCE),” HESS. 45	Closely relevant global weighted ET synthesis; combines products using observationally constrained weights.
Hobeichi et al. (2021), “Robust historical evapotranspiration trends across climate regimes,” HESS. 12	Particularly useful for ITHACA's spatial grouping choice and the trade-off between regime specificity and sample size.
Reitz et al. (2025), “Performance Mapping and Weighting for the Evapotranspiration Models of the OpenET Ensemble,” Water Resources Research. 11	Probably the most directly useful recent reference for spatially varying model performance, alternative metric-to-weight transforms and weighting-strength sensitivity.
Giorgi & Mearns (2002), Reliability Ensemble Averaging, Journal of Climate. 46	Classic precedent for converting performance and agreement information into ensemble reliability weights.
Raftery et al. (2005), “Using Bayesian Model Averaging to Calibrate Forecast Ensembles,” Monthly Weather Review. 26	Defines what genuinely probabilistic/Bayesian mixture weights mean; useful for precisely distinguishing ITHACA's selection probabilities from posterior model probabilities.

Reference	Core methodological role for ITHACA
Weigel et al. (2010), "Risks of Model Weighting in Multimodel Climate Projections," <i>Journal of Climate</i>. ¹⁵	Essential counterweight: erroneous skill weights can be worse than equal weighting. Strong justification for ITHACA's neutral scenario.
Knutti et al. (2017), "A climate model projection weighting scheme accounting for performance and interdependence," <i>GRL</i>. ¹⁴	Key support for considering both performance and member dependence; directly relevant to ITHACA's remaining genealogy problem.
Yang et al. (2008), "New analytical derivation of the mean annual water–energy balance equation," <i>Water Resources Research</i>. ¹⁶	Strong theoretical foundation for long-term P–E–energy plausibility screening.
Greve et al. (2016), "A two-parameter Budyko function to represent conditions under which evapotranspiration exceeds precipitation," <i>HESS</i>. ¹⁸	Essential caution against treating $E \leq P$ as a universal hard constraint.
Pan et al. (2012), "Multisource Estimation of Long-Term Terrestrial Water Budget for Major Global River Basins," <i>Journal of Climate</i>. ⁴⁷	Major precedent for multisource P/E/water-budget synthesis and physical closure constraints.
Aires (2014), "Combining Datasets of Satellite-Retrieved Products. Part I: Methodology and Water Budget Closure," <i>Journal of Hydrometeorology</i>. ⁴⁸	Direct methodological precedent for multi-variable observation fusion and water-budget consistency.
Schoups & Nasser (2021), "GRACEfully Closing the Water Balance," <i>Water Resources Research</i>. ⁴⁹	Important probabilistic precedent for joint inference of multiple hydrological variables and correlated uncertainty.
Abolafia-Rosenzweig et al. (2020), remotely sensed terrestrial water-budget ensembles, <i>Remote Sensing of Environment</i>. ²¹	Especially relevant precedent for retaining multiple alternative water-budget realizations formed from different source-dataset choices rather than one deterministic optimum.
Cornes et al. (2018), "An Ensemble Version of the E-OBS Temperature and Precipitation Data Sets," <i>JGR Atmospheres</i>. ²⁰	Important observational-ensemble precedent: uncertainty represented by stochastic members rather than only a deterministic field plus error bars.
Mourad et al. (2026), joint probabilistic water-balance data fusion, <i>HESS</i>. ⁵⁰	Current-frontier evidence for jointly calibrating hydrological variables and allowing physical closure to generate multivariate dependence rather than handling fluxes independently.

The resulting methodological citation architecture should make one distinction repeatedly and consistently. **ITHACA is not trying to estimate a unique optimal fused P or E field.** Studies such as DOLCE, SUPER, REA/BMA and water-budget-constrained fusion establish that product skill can be learned, weighted and physically constrained. ⁵¹ ITHACA instead uses such evaluation information to define a categorical

distribution over predefined P-E worlds and propagates that discrete structural choice through Monte Carlo sampling while retaining source identity.

That methodological position is defensible **provided the manuscript is explicit that the ensemble is conditional and structurally probabilistic rather than statistically calibrated to the unknown truth.** The strongest version of the Methods would therefore describe the pipeline in four epistemic layers: **consensus evidence → plausibility screening → relative selection weights → provenance-preserving structural sampling.** The literature supports each layer individually; what requires empirical demonstration in ITHACA is the robustness of the transitions between them—especially the reference composition, trend-significance score, weighting transform and physical-screen thresholds.

1 26 Using Bayesian Model Averaging to Calibrate Forecast Ensembles

<https://doi.org/10.1175%2FMWR2906.1>

2 8 A modified Mann-Kendall trend test for autocorrelated data - ScienceDirect

<https://doi.org/10.1016/s0022-1694%2897%2900125-x>

3 31 34 39 Toward the true near-surface wind speed: Error modeling and calibration using triple collocation - Stoffelen - 1998 - Journal of Geophysical Research: Oceans - Wiley Online Library

<https://doi.org/10.1029/97JC03180>

4 25 42 Statistical uncertainty analysis-based precipitation merging (SUPER): A new framework for improved global precipitation estimation - ScienceDirect

<https://www.sciencedirect.com/science/article/pii/S0034425722004059>

5 38 41 Evaluation of Assumptions in Soil Moisture Triple Collocation Analysis in: Journal of Hydrometeorology Volume 15 Issue 3 (2014)

https://doi.org/10.1175/JHM-D-13-0158.1?urlappend=%3Futm_source%3Dresearchgate.net%26utm_medium%3Darticle

6 43 Summarizing multiple aspects of model performance in a single diagram - Taylor - 2001 - Journal of Geophysical Research: Atmospheres - Wiley Online Library

<https://doi.org/10.1029/2000jd900719>

7 Estimates of the Regression Coefficient Based on Kendall's Tau | JSTOR

<https://www.jstor.org/stable/2285891?origin=crossref>

9 29 30 HESS - The WACMOS-ET project – Part 2: Evaluation of global terrestrial evaporation data sets

<https://hess.copernicus.org/articles/20/823/2016/>

10 45 51 HESS - Relations - Derived Optimal Linear Combination Evapotranspiration (DOLCE): a global gridded synthesis ET estimate

<https://hess.copernicus.org/articles/22/1317/2018/hess-22-1317-2018-relations.html>

11 Performance Mapping and Weighting for the Evapotranspiration Models of the OpenET Ensemble - Reitz - 2025 - Water Resources Research - Wiley Online Library

<https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2024WR038899>

12 HESS - Robust historical evapotranspiration trends across climate regimes

<https://hess.copernicus.org/articles/25/3855/2021/>

- 13 35 46 Calculation of Average, Uncertainty Range, and Reliability of Regional Climate Changes from AOGCM Simulations via the "Reliability Ensemble Averaging" (REA) Method
https://journals.ametsoc.org/abstract/journals/clim/15/10/1520-0442_2002_015_1141_coaura_2.0.co_2.xml
- 14 A climate model projection weighting scheme accounting for performance and interdependence - Knutti - 2017 - Geophysical Research Letters - Wiley Online Library
<https://agupubs.onlinelibrary.wiley.com/doi/full/10.1002/2016GL072012>
- 15 23 28 32 Risks of Model Weighting in Multimodel Climate Projections
<https://journals.ametsoc.org/abstract/journals/clim/23/15/2010jcli3594.1.xml>
- 16 27 36 New analytical derivation of the mean annual water-energy balance equation - Yang - 2008 - Water Resources Research - Wiley Online Library
<https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2007wr006135>
- 17 47 Multisource Estimation of Long-Term Terrestrial Water Budget for Major Global River Basins
<https://journals.ametsoc.org/view/journals/clim/25/9/jcli-d-11-00300.1.xml>
- 18 HESS - A two-parameter Budyko function to represent conditions under which evapotranspiration exceeds precipitation
<https://hess.copernicus.org/articles/20/2195/2016/>
- 19 49 GRACEfully Closing the Water Balance: A Data-Driven Probabilistic Approach Applied to River Basins in Iran - Schoups - 2021 - Water Resources Research - Wiley Online Library
<https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2020WR029071>
- 20 33 An Ensemble Version of the E-OBS Temperature and Precipitation Data Sets - Cornes - 2018 - Journal of Geophysical Research: Atmospheres - Wiley Online Library
<https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2017JD028200>
- 21 Remotely sensed ensembles of the terrestrial water budget over major global river basins: An assessment of three closure techniques - ScienceDirect
<https://www.sciencedirect.com/science/article/pii/S0034425720305642>
- 22 On the Effectiveness of Common Random Numbers | Management Science
<https://pubsonline.informs.org/doi/10.1287/mnsc.25.7.649>
- 24 Error characterization of global land evapotranspiration products: Collocation-based approach - ScienceDirect
<https://www.sciencedirect.com/science/article/pii/S0022169422006771>
- 37 40 Extended triple collocation: Estimating errors and correlation coefficients with respect to an unknown target - McColl - 2014 - Geophysical Research Letters - Wiley Online Library
<https://agupubs.onlinelibrary.wiley.com/doi/10.1002/2014GL061322>
- 44 Performance metrics for climate models - Gleckler - 2008 - Journal of Geophysical Research: Atmospheres - Wiley Online Library
<https://doi.org/10.1029/2007jd008972>
- 48 Combining Datasets of Satellite-Retrieved Products. Part I: Methodology and Water Budget Closure
https://journals.ametsoc.org/view/journals/hydr/15/4/jhm-d-13-0148_1.xml
- 50 HESS - Joint calibration of multi-scale hydrological data sets using probabilistic water balance data fusion: methodology and application to the irrigated Hindon River Basin, India
<https://hess.copernicus.org/articles/30/525/2026/>